MECH 223 Solved Examples — Chapter 6 (Trusses)

MECH 223-002 Engineering Statics, Fall 2026
Chapter 6: simple planar trusses

Set B

How to use these. Try each example yourself after reading only *Given/Find* and the figure. Then compare with the Strategy and Solution. Members are two-force members, joints are frictionless pins, and loads act at joints. Sign convention: a member force is **tension (T) when positive** (the member pulls on the joint) and **compression (C) when negative** (the member pushes on the joint). Every answer states T or C with a magnitude. For whole-truss moments, counterclockwise (CCW) is positive. Numbers are new.

1 Method of joints: a triangular truss with a sideways load

Level: easy. Chapter 6. Target confusion: finding reactions first, then starting at the joint that has only two unknown members.

1. Given / Find

A triangular truss ABC has a pin at A and a roller at C (the roller force is vertical). Joints A and C are 6.00 m apart on a horizontal line, and joint B is 4.00 m above the midpoint of AC, so $AB = BC = 5.00$ m. At B the truss carries a 12.0 kN downward load and a 6.00 kN horizontal load pointing right.

Find: the support reactions, then the forces in members AB and AC by the method of joints at A. Finish with member BC from joint C, and confirm with joint B. State T or C for each.

2. Figure

3. Free-body diagrams

4. Strategy

First treat the whole truss as one rigid body and find the three reactions. The joint at A has two unknown members (AB and AC) and two known reaction components, so it can be solved with $\sum F_x = 0$ and $\sum F_y = 0$. Assume every member is in tension, so the force arrow points away from the joint. A negative answer then means compression. Unit vectors along members come from the 3-4-5 triangle: from A toward B the direction is (3/5, 4/5).

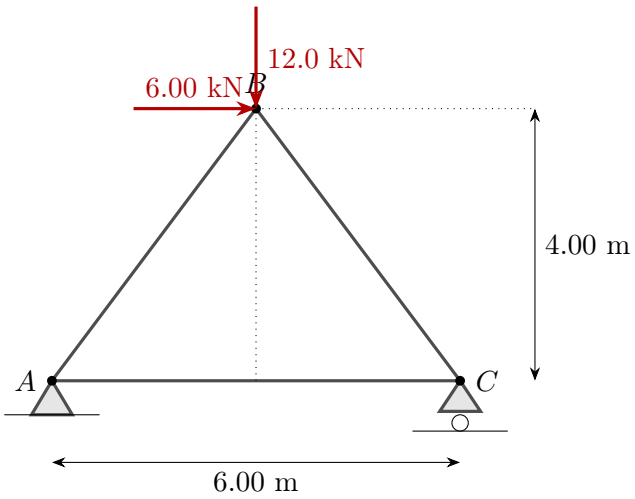


Figure 1: Triangular truss with a pin at A and a roller at C. Joint B is 3.00 m right of A and 4.00 m above the line AC. A 12.0 kN load acts downward and a 6.00 kN load acts to the right, both at B.

5. Solution

Reactions. Take moments about A (CCW positive). The force 12.0 kN down at B has lever arm 3.00 m and the 6.00 kN force to the right acts 4.00 m above A. Both turn the truss clockwise:

$$\sum M_A = C_y(6.00) - 12.0(3.00) - 6.00(4.00) = 0$$
$$C_y = \frac{36.0 + 24.0}{6.00} = 10.0 \text{ kN (up)}.$$

$$\sum F_y = A_y + C_y - 12.0 = 0 \;\Rightarrow\; A_y = 2.00 \text{ kN (up)},$$
$$\sum F_x = A_x + 6.00 = 0 \;\Rightarrow\; A_x = -6.00 \text{ kN, i.e. } 6.00 \text{ kN to the left.}$$

Joint A. With tension assumed, F_{AB} acts along (3/5, 4/5) and F_{AC} acts along (1, 0).

$$\sum F_y = 2.00 + \tfrac{4}{5}F_{AB} = 0 \;\Rightarrow\; F_{AB} = -2.50 \text{ kN},$$
$$\sum F_x = -6.00 + \tfrac{3}{5}F_{AB} + F_{AC} = 0 \;\Rightarrow\; F_{AC} = 6.00 + 1.50 = 7.50 \text{ kN}.$$

So AB is in compression and AC is in tension.

Joint C (member BC). From C toward B the direction is $(-3/5,\ 4/5)$, and $C_y = 10.0$ kN acts up.

$$\sum F_y = 10.0 + \tfrac{4}{5}F_{BC} = 0 \;\Rightarrow\; F_{BC} = -12.5 \text{ kN},$$
$$\sum F_x = -\tfrac{3}{5}F_{BC} - F_{AC} = 0 \;\Rightarrow\; F_{AC} = 7.50 \text{ kN}.$$

The second equation gives the same F_{AC} found at joint A, so the two joints agree.

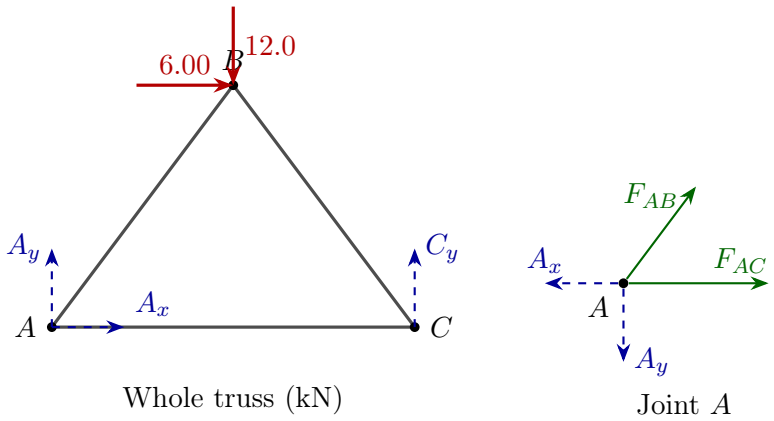


Figure 2: Left: free-body diagram of the whole truss with the unknown reactions A_x, A_y and C_y drawn as blue dashed arrows. Right: free-body diagram of joint A. The green arrows are the member forces F_{AB} and F_{AC}, both drawn pulling away from the joint (assumed tension). The reaction arrows at A are drawn in the directions found in the solution, A_x to the left and A_y upward.

Joint B as a check. The member forces on B are F_{AB} along $(-3/5, -4/5)$ and F_{BC} along $(3/5, -4/5)$, plus the two applied loads:

$$\sum F_x = (-2.50)(-0.6) + (-12.5)(0.6) + 6.00 = 1.50 - 7.50 + 6.00 = 0,$$
$$\sum F_y = (-2.50)(-0.8) + (-12.5)(-0.8) - 12.0 = 2.00 + 10.0 - 12.0 = 0.$$

6. Answer

$$\boxed{F_{AB} = 2.50 \text{ kN (C)}}, \qquad \boxed{F_{AC} = 7.50 \text{ kN (T)}}, \qquad \boxed{F_{BC} = 12.5 \text{ kN (C)}}$$

Reactions: $A_x = 6.00$ kN (left), $A_y = 2.00$ kN (up), $C_y = 10.0$ kN (up).

7. Sanity check

Joint B balances in both directions, which is an equation set that was not used to find any of the three forces. The bottom member AC is in tension and the two inclined members are in compression, as expected for a triangle loaded from above with a push at the apex. The right inclined member carries the most force because the roller at C takes most of the vertical load (10.0 of 12.0 kN).

2 Zero-force members by inspection, then joints

Level: easy to medium. Chapter 6. Target confusion: a zero-force member is not a useless member. It is a member that carries no load for this particular loading.

1. Given / Find

A truss has a pin at A and a roller at C (vertical reaction). Joints A, B and C lie on a horizontal line with $AB = BC = 4.00$ m. Joint D is 3.00 m directly above B. Joint E is the midpoint of AD, so AE and ED are collinear. Members are AB, BC, AE, ED, DC, BD and BE. At D the truss carries a 12.0 kN downward load and an 8.00 kN horizontal load pointing right. Nothing is applied at E or B.

Find: (a) which members carry zero force, using inspection only; (b) the reactions; (c) the forces in AE, ED, AB, BC and DC. State T or C for each.

2. Figure

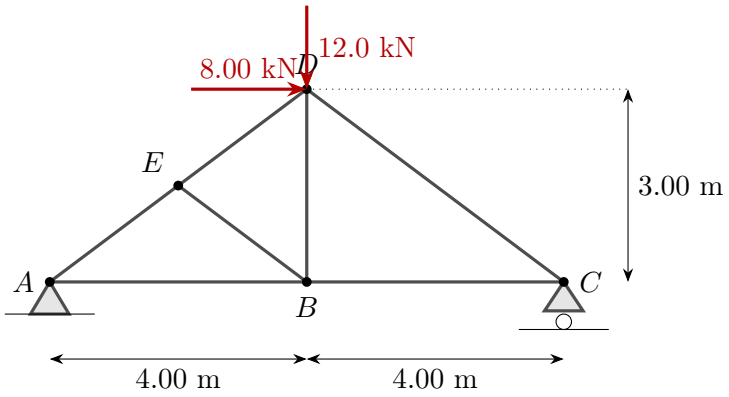


Figure 3: Truss with joints A, B, C on a horizontal line, apex D directly above B, and joint E at the midpoint of AD. Member BE joins B to E. A pin supports A and a roller supports C. At D a 12.0 kN load acts downward and an 8.00 kN load acts to the right.

3. Free-body diagrams

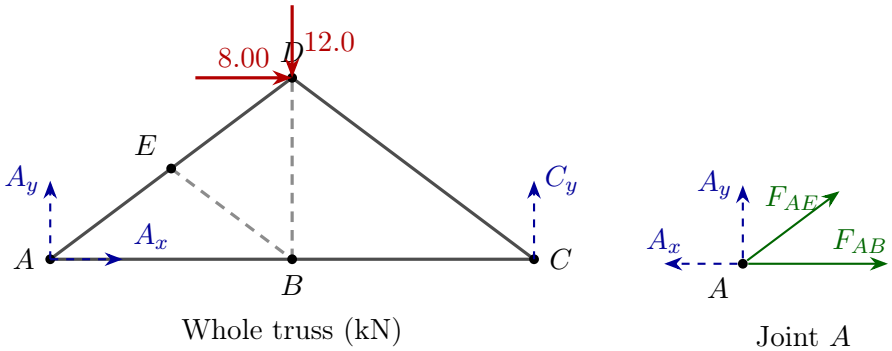


Figure 4: Left: free-body diagram of the whole truss with the unknown reactions as blue dashed arrows. The two members found to be zero-force by inspection, BE and BD, are drawn as grey dashed lines. Right: joint A with member forces drawn in tension (green) and the reactions in the directions found in the solution (A_x to the left, A_y upward).

4. Strategy

Two inspection rules do most of the work. (i) At an unloaded joint where three members meet and two of them are collinear, the third member carries zero force. (ii) At an unloaded joint where only two non-collinear members meet, both carry zero force. Apply them in order, because removing one zero-force member can create another. After that, solve the reactions on the whole truss and the remaining forces with joints A and C.

5. Solution

(a) Zero-force members.

- *Joint E.* No load acts at E. Members AE and ED lie on the same line, so the component of forces perpendicular to that line comes only from BE. Hence $F_{BE} = 0$, and $F_{AE} = F_{ED}$ because the two collinear forces must balance.
- *Joint B.* No load acts at B. Members AB and BC are collinear horizontal members, and BE is now known to be zero. The only vertical force on B is F_{BD}, so $\sum F_y = 0$ gives $F_{BD} = 0$. Since AB and BC are then the only members carrying force at B, $\sum F_x = 0$ gives $F_{AB} = F_{BC}$.

The zero-force members are BE and BD. They are not removed from the structure, since they keep B and E in position and would carry load if the loading changed.

(b) Reactions. Moments about A, CCW positive. The load at D has lever arm 4.00 m for the vertical part and 3.00 m for the horizontal part:

$$\sum M_A = C_y(8.00) - 12.0(4.00) - 8.00(3.00) = 0$$
$$C_y = \frac{48.0 + 24.0}{8.00} = 9.00 \text{ kN (up)}.$$

$$\sum F_y = A_y + 9.00 - 12.0 = 0 \;\Rightarrow\; A_y = 3.00 \text{ kN (up)},$$
$$\sum F_x = A_x + 8.00 = 0 \;\Rightarrow\; A_x = -8.00 \text{ kN, i.e. } 8.00 \text{ kN to the left.}$$

(c) Joint A. Member AE rises 1.50 m over a run of 2.00 m with length 2.50 m, so its direction from A is (0.8, 0.6). Member AB points along (1, 0).

$$\sum F_y = 3.00 + 0.6\,F_{AE} = 0 \;\Rightarrow\; F_{AE} = -5.00 \text{ kN},$$
$$\sum F_x = -8.00 + 0.8\,F_{AE} + F_{AB} = 0 \;\Rightarrow\; F_{AB} = 8.00 + 4.00 = 12.0 \text{ kN}.$$

From (a), $F_{ED} = F_{AE} = -5.00$ kN and $F_{BC} = F_{AB} = 12.0$ kN.

Joint C. Member CD points from C along $(-0.8, 0.6)$, and $C_y = 9.00$ kN acts up.

$$\sum F_y = 9.00 + 0.6\,F_{CD} = 0 \;\Rightarrow\; F_{CD} = -15.0 \text{ kN},$$
$$\sum F_x = -0.8\,F_{CD} - F_{BC} = 12.0 - F_{BC} = 0 \;\Rightarrow\; F_{BC} = 12.0 \text{ kN}.$$

This matches F_{BC} from joint A.

6. Answer

$\boxed{F_{BE} = F_{BD} = 0}$, $\boxed{F_{AE} = F_{ED} = 5.00 \text{ kN (C)}}$, $\boxed{F_{AB} = F_{BC} = 12.0 \text{ kN (T)}}$, $\boxed{F_{CD} = 15.0 \text{ kN (C)}}$

Reactions: $A_x = 8.00$ kN (left), $A_y = 3.00$ kN (up), $C_y = 9.00$ kN (up).

7. Sanity check

Joint D was not used to find any force, so it makes a clean test. The member forces on D are F_{DE} along $(-0.8, -0.6)$, F_{DC} along $(0.8, -0.6)$ and $F_{DB} = 0$, plus the applied loads:

$$\sum F_x = (-5.00)(-0.8) + (-15.0)(0.8) + 8.00 = 4.00 - 12.0 + 8.00 = 0,$$
$$\sum F_y = (-5.00)(-0.6) + (-15.0)(-0.6) - 12.0 = 3.00 + 9.00 - 12.0 = 0.$$

Both balance. With BE and BD removed, the truss behaves as a plain triangle ADC with a loaded apex, which explains why the two inclined members are in compression and the bottom chord is in tension.

3 Method of joints through several joints

Level: medium. Chapter 6. Target confusion: choosing the next joint. Each joint is only solvable when it has at most two unknown member forces, so the order matters.

1. Given / Find

A Warren truss has bottom joints A, C, E and top joints B, D. The joints A, C, E lie on a horizontal line at $x = 0$, 6.00 and 12.00 m. The top joints B and D are 4.00 m above the line, at $x = 3.00$ and 9.00 m, so every inclined member is 5.00 m long (a 3-4-5 triangle). The members are AB, BD, DE, AC, CE, BC and CD. A pin supports A and a roller supports E (vertical reaction). Downward loads act at three joints: 10.0 kN at B, 6.00 kN at C and 6.00 kN at D.

Find: the support reactions, then the force in every member by working from joint to joint. Finish with a table of member forces, each marked T or C.

2. Figure

3. Free-body diagrams

4. Strategy

Find the three reactions from the whole truss. Then pick a joint with at most two unknown member forces. Joint A qualifies, because it has the reaction A_y and only members AB and AC. Solving it makes F_{AB} and F_{AC} known, which leaves joint B with only two unknowns (BC and BD), then joint C with two unknowns (CD and CE), then joint E with one unknown (DE). The last joint, D, uses none of its equations to find a force, so it becomes the check. Assume tension everywhere

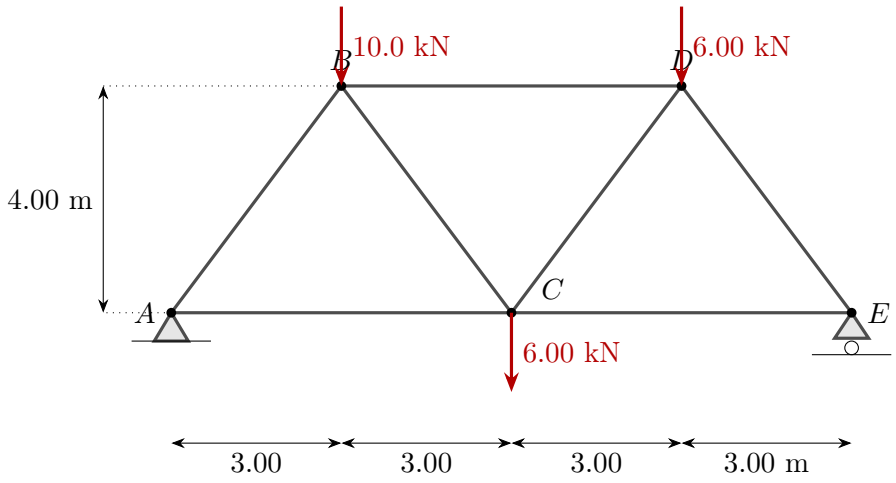
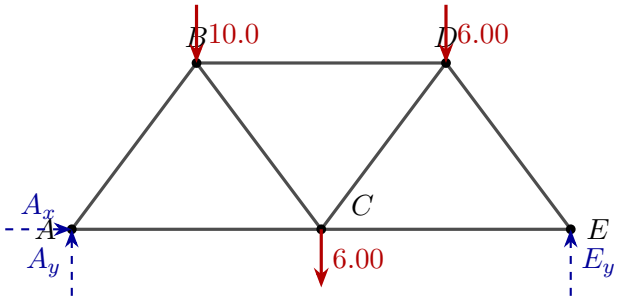


Figure 5: Warren truss with a pin at A and a roller at E. Horizontal spacing between neighbouring joints is 3.00 m and the top joints are 4.00 m above the bottom chord. Downward loads: 10.0 kN at B, 6.00 kN at C, 6.00 kN at D.



Whole truss (kN)

Figure 6: Free-body diagram of the whole truss. The pin at A supplies A_x and A_y and the roller at E supplies E_y, all drawn as blue dashed arrows. The solution gives $A_x = 0$, $A_y = 12.0$ kN and $E_y = 10.0$ kN, both upward.

(arrow away from the joint). Unit vectors: an inclined member rising to the right has direction $(0.6,\ 0.8)$, and one falling to the right has direction $(0.6,\ -0.8)$.

5. Solution

Reactions. Moments about A (CCW positive). All three loads are downward and produce clockwise moments:

$$\sum M_A = E_y(12.0) - 10.0(3.00) - 6.00(6.00) - 6.00(9.00) = 0$$

$$E_y = \frac{30.0 + 36.0 + 54.0}{12.0} = 10.0 \text{ kN (up)}.$$

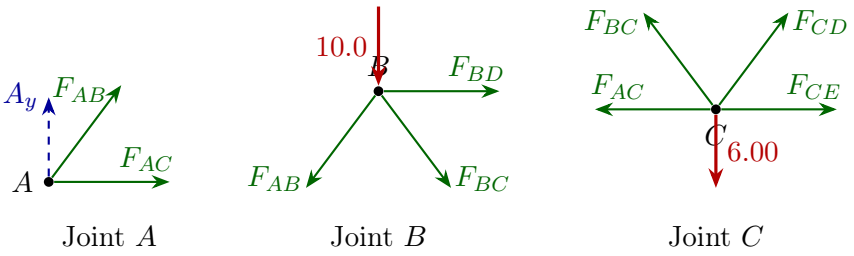


Figure 7: Free-body diagrams of joints A, B and C (loads in kN). Every member force is drawn pulling away from the joint, which assumes tension. Joint A shows its reaction A_y in the direction found in the solution; $A_x = 0$ so it is not drawn.

$$\sum F_y = A_y + 10.0 - 22.0 = 0 \;\Rightarrow\; A_y = 12.0 \text{ kN (up)},$$
$$\sum F_x = A_x = 0.$$

Joint A (unknowns F_{AB}, F_{AC}). Member AB points along (0.6, 0.8) and AC along (1, 0).

$$\sum F_y = 12.0 + 0.8\,F_{AB} = 0 \;\Rightarrow\; F_{AB} = -15.0 \text{ kN},$$
$$\sum F_x = 0.6\,F_{AB} + F_{AC} = 0 \;\Rightarrow\; F_{AC} = 9.00 \text{ kN}.$$

AB is in compression and AC is in tension.

Joint B (unknowns F_{BC}, F_{BD}). From B, member BA points along $(-0.6, -0.8)$, BC along $(0.6, -0.8)$ and BD along (1, 0). The 10.0 kN load acts downward.

$$\sum F_y = -0.8(-15.0) - 0.8\,F_{BC} - 10.0 = 12.0 - 10.0 - 0.8\,F_{BC} = 0 \;\Rightarrow\; F_{BC} = 2.50 \text{ kN},$$
$$\sum F_x = -0.6(-15.0) + 0.6(2.50) + F_{BD} = 9.00 + 1.50 + F_{BD} = 0 \;\Rightarrow\; F_{BD} = -10.5 \text{ kN}.$$

BC is in tension and BD is in compression.

Joint C (unknowns F_{CD}, F_{CE}). From C, member CA points along $(-1, 0)$, CB along $(-0.6, 0.8)$, CD along (0.6, 0.8) and CE along (1, 0). The 6.00 kN load acts downward.

$$\sum F_y = 0.8(2.50) + 0.8\,F_{CD} - 6.00 = 0 \;\Rightarrow\; F_{CD} = \frac{6.00 - 2.00}{0.8} = 5.00 \text{ kN},$$
$$\sum F_x = -9.00 - 0.6(2.50) + 0.6\,F_{CD} + F_{CE} = -9.00 - 1.50 + 3.00 + F_{CE} = 0 \;\Rightarrow\; F_{CE} = 7.50 \text{ kN}.$$

Both CD and CE are in tension.

Joint E (unknown F_{DE}, with F_{CE} already known). From E, member ED points along $(-0.6, 0.8)$ and EC along $(-1, 0)$. The reaction $E_y = 10.0$ kN acts up.

$$\sum F_y = 10.0 + 0.8\,F_{DE} = 0 \;\Rightarrow\; F_{DE} = -12.5 \text{ kN}.$$

DE is in compression. The x equation was not needed, so it checks the work: $-0.6(-12.5) - F_{CE} = 7.50 - 7.50 = 0$.

Member force table (tension positive):

Member	Force (kN)	Magnitude (kN)	T or C
AB	-15.0	15.0	C
BD	-10.5	10.5	C
DE	-12.5	12.5	C
AC	$+9.00$	9.00	T
CE	$+7.50$	7.50	T
BC	$+2.50$	2.50	T
CD	$+5.00$	5.00	T

6. Answer

$$\boxed{F_{AB} = 15.0 \text{ kN (C)}},\ \boxed{F_{BD} = 10.5 \text{ kN (C)}},\ \boxed{F_{DE} = 12.5 \text{ kN (C)}},$$

$$\boxed{F_{AC} = 9.00 \text{ kN (T)}},\ \boxed{F_{CE} = 7.50 \text{ kN (T)}},\ \boxed{F_{BC} = 2.50 \text{ kN (T)}},\ \boxed{F_{CD} = 5.00 \text{ kN (T)}}$$

Reactions: $A_x = 0$, $A_y = 12.0$ kN (up), $E_y = 10.0$ kN (up).

7. Sanity check

Joint D was never used to find a force. The member forces on D are $F_{DB} = -10.5$ along $(-1, 0)$, $F_{DC} = 5.00$ along $(-0.6, -0.8)$ and $F_{DE} = -12.5$ along $(0.6, -0.8)$, with the 6.00 kN load downward:

$$\sum F_x = (-10.5)(-1) + 5.00(-0.6) + (-12.5)(0.6) = 10.5 - 3.00 - 7.50 = 0,$$
$$\sum F_y = 5.00(-0.8) + (-12.5)(-0.8) - 6.00 = -4.00 + 10.0 - 6.00 = 0.$$

Both balance. The pattern is also physical: the top chord and the two end members are in compression and the bottom chord is in tension, as for a beam loaded downward. The two end members carry the largest forces because they pass the full reaction into the supports ($12.0/0.8 = 15.0$ and $10.0/0.8 = 12.5$ kN).

4 Method of sections: cutting three members

Level: medium to hard. Chapter 6. Target confusion: where to put the moment center. Pick the point where two of the three cut members cross, and the third force is the only unknown left in that moment equation.

1. Given / Find

A Pratt truss has bottom joints A, B, C, D, E on a horizontal line, spaced 3.00 m apart. The top joints F, G, H lie 4.00 m directly above B, C, D. The members are the bottom chord AB, BC, CD, DE, the top chord FG, GH, the end posts AF and HE, the verticals BF, CG, DH, and the diagonals FC and HC. A pin supports A and a roller supports E (vertical reaction). Downward loads act on the bottom joints: 8.00 kN at B, 12.0 kN at C and 4.00 kN at D.

Find: the reactions, then the forces in members FG, FC and BC by the method of sections, using one section through the panel between B and C. Do not solve the truss joint by joint. State T or C for each member.

2. Figure

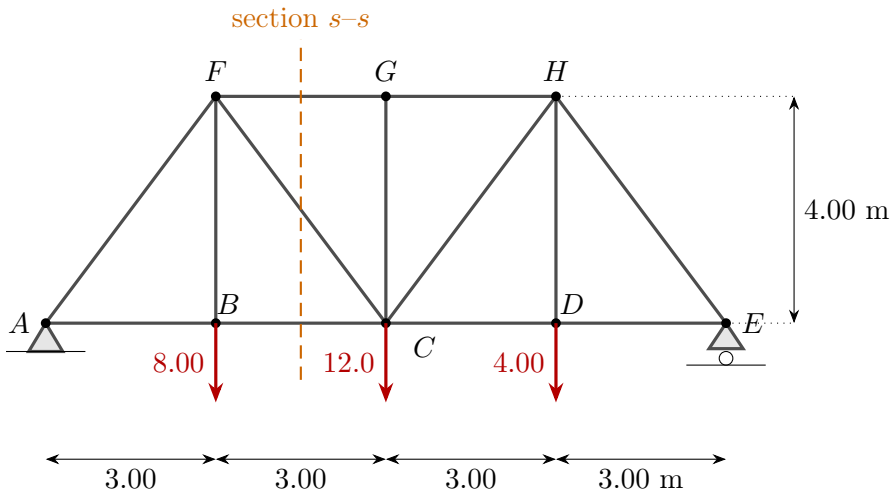


Figure 8: Pratt truss with a pin at A and a roller at E. Loads in kN act downward at B, C and D. The orange dashed line marks section s–s, which passes through the three members FG, FC and BC only.

3. Free-body diagrams

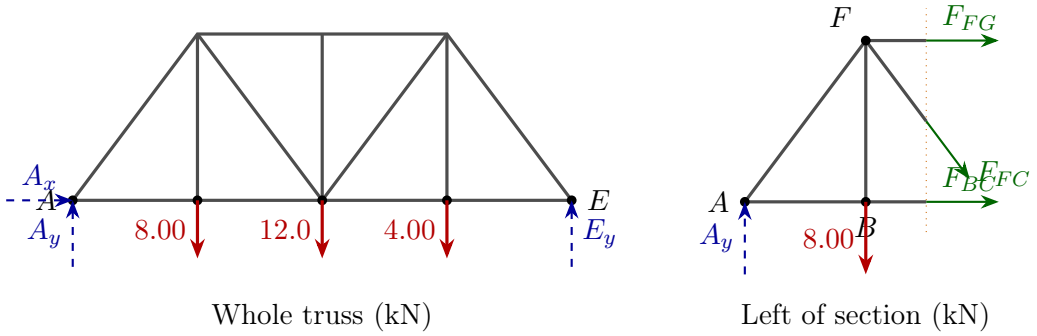


Figure 9: Left: whole-truss free-body diagram with A_x, A_y, E_y as blue dashed arrows. Right: the part of the truss to the left of section s–s. It contains A, B, F and members AB, AF, BF, plus the cut ends of FG, FC and BC. The three cut-member forces (green) are drawn pulling away from the body, which assumes tension. $A_y = 13.0$ kN upward comes from the whole-truss solution and $A_x = 0$.

4. Strategy

Find the reactions first. Then pass section s–s through the panel between B and C. It cuts exactly three members: the top chord FG, the diagonal FC and the bottom chord BC. Keep the left part. Three independent equations are available, and the best choice is a moment center that kills two unknowns at once:

- About C: members FC and BC both pass through C, so only F_{FG} appears.
- About F: members FG and FC both pass through F, so only F_{BC} appears.
- $\sum F_y = 0$: the two chords are horizontal, so only the diagonal enters, giving F_{FC}.

The remaining equation, $\sum F_x = 0$, is saved for the check. The diagonal runs from $F(3, 4)$ to $C(6, 0)$, so its direction from F is (0.6, −0.8).

5. Solution

Reactions. Moments about A (CCW positive), loads at $x = 3.00$, 6.00, 9.00 m:

$$\sum M_A = E_y(12.0) - 8.00(3.00) - 12.0(6.00) - 4.00(9.00) = 0$$
$$E_y = \frac{24.0 + 72.0 + 36.0}{12.0} = 11.0 \text{ kN (up)}.$$

$$\sum F_y = A_y + 11.0 - 24.0 = 0 \;\Rightarrow\; A_y = 13.0 \text{ kN (up)},$$
$$\sum F_x = A_x = 0.$$

Section, left part. Forces on the left part: $A_y = 13.0$ kN up at A, 8.00 kN down at B, and the three cut forces, assumed tension. F_{FG} and F_{BC} act in the $+x$ direction on the left part, and F_{FC} acts along (0.6, −0.8).

Moments about C (6, 0) eliminate F_{FC} and F_{BC}. The top-chord force acts 4.00 m above C and pulls to the right, which gives a clockwise moment for a positive (tensile) value:

$$\sum M_C = -13.0(6.00) + 8.00(3.00) - F_{FG}(4.00) = 0$$
$$-78.0 + 24.0 = 4.00\,F_{FG} \;\Rightarrow\; F_{FG} = -13.5 \text{ kN}.$$

Moments about F (3, 4) eliminate F_{FG} and F_{FC}. The 8.00 kN load at B passes straight below F and has zero lever arm. The bottom-chord force acts 4.00 m below F, and a positive value pulls right, which gives a counterclockwise moment:

$$\sum M_F = -13.0(3.00) + 0 + F_{BC}(4.00) = 0$$
$$F_{BC} = \frac{39.0}{4.00} = 9.75 \text{ kN}.$$

Vertical forces. Only A_y, the 8.00 kN load and the vertical part of F_{FC} act vertically:

$$\sum F_y = 13.0 - 8.00 - 0.8\,F_{FC} = 0 \;\Rightarrow\; F_{FC} = \frac{5.00}{0.8} = 6.25 \text{ kN}.$$

6. Answer

$$\boxed{F_{FG} = 13.5 \text{ kN (C)}}, \qquad \boxed{F_{FC} = 6.25 \text{ kN (T)}}, \qquad \boxed{F_{BC} = 9.75 \text{ kN (T)}}$$

Reactions: $A_x = 0$, $A_y = 13.0$ kN (up), $E_y = 11.0$ kN (up).

7. Sanity check

Horizontal balance (unused equation). On the left part,

$$\sum F_x = F_{FG} + F_{BC} + 0.6\, F_{FC} = -13.5 + 9.75 + 3.75 = 0.$$

Right-hand part. Take moments about C for the part to the right of the section, which contains C, D, E, G, H. The cut forces of FC and BC pass through C, and the 12.0 kN load at C has zero lever arm. The top-chord cut force on this part points left when F_{FG} is a tension, so it acts 4.00 m above C and pulls to the left, which is a counterclockwise moment of $+4.00\, F_{FG}$:

$$\sum M_C = 11.0(6.00) - 4.00(3.00) + 4.00\, F_{FG} = 66.0 - 12.0 + 4.00\, F_{FG} = 0 \;\Rightarrow\; F_{FG} = -13.5 \text{ kN},$$

the same compressive value found from the left part. The pattern is physical: the top chord is squeezed and the bottom chord is stretched, as in a beam bending under downward loads, and the diagonal in the left panel is in tension, as expected for a Pratt truss.

5 Method of sections for a member far from the supports

Level: hard. Chapter 6. Target confusion: believing you must work in from the supports. A member in the middle of a long truss needs one cut and one equation, because the cut can be placed right next to it.

1. Given / Find

A Pratt truss has six equal panels, each 1.80 m wide. The bottom joints A, B, C, D, E, F, G lie on a horizontal line. The top joints H, I, J, K, L lie 2.40 m directly above B, C, D, E, F. The members are the bottom chord AB, BC, CD, DE, EF, FG, the top chord HI, IJ, JK, KL, the end posts AH and GL, the verticals BH, CI, DJ, EK, FL, and the diagonals HC, ID, KD, LE. A pin supports A and a roller supports G (vertical reaction). Downward loads act on the bottom joints: 2.00 kN at B, 9.00 kN at C, 11.0 kN at D, 4.00 kN at E and 3.00 kN at F.

Find: the reactions, then the force in the single diagonal ID, using one section and one equation. State T or C. Also state how many joints you would have had to solve to reach ID without a section.

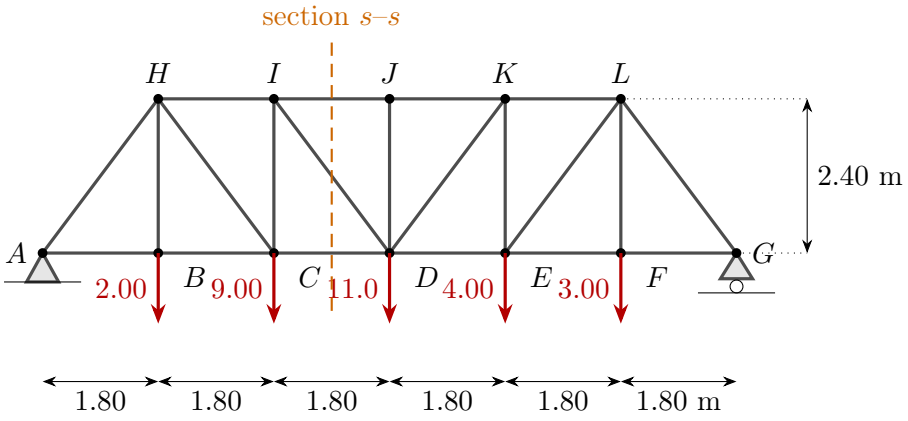
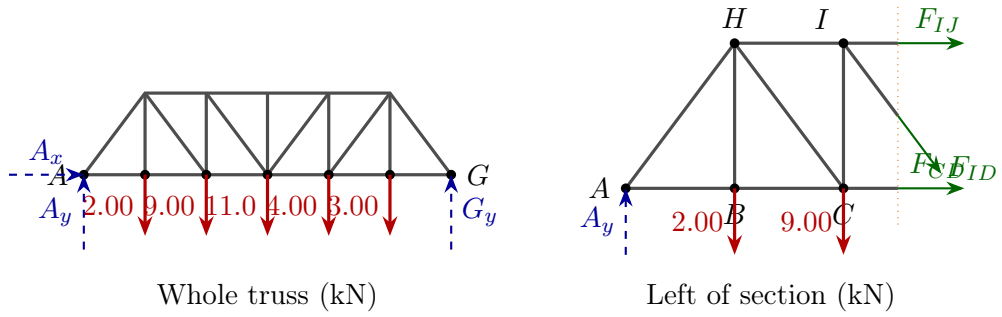


Figure 10: Six-panel Pratt truss with a pin at A and a roller at G. Loads in kN act downward at B to F. The orange dashed line marks section s–s, which lies in the panel between C and D and passes through only IJ, ID and CD. The diagonal ID is three panels from A and is the member to find.



Whole truss (kN) Left of section (kN)

Figure 11: Left: whole-truss free-body diagram with the unknown reactions A_x, A_y, G_y as blue dashed arrows. Right: the part of the truss to the left of section s–s. It contains A, B, C, H, I and the members between them, plus the cut ends of IJ, ID and CD. The three cut-member forces (green) are drawn pulling away from the body, which assumes tension. $A_y = 15.0$ kN upward comes from the whole-truss solution.

2. Figure

3. Free-body diagrams

4. Strategy

Find the reactions first. Then place section s–s in the panel between C and D, directly beside the diagonal ID. A vertical line there crosses exactly three members: the top chord IJ, the diagonal ID and the bottom chord CD. Keep the left part; it has only A_y and two loads on it.

Both chords are horizontal, so neither contributes to a vertical force balance. That leaves one unknown in one equation:

- $\sum F_y = 0$ on the left part involves only A_y, the loads at B and C, and the vertical component of F_{ID}.

The diagonal runs from $I(3.60,\ 2.40)$ to $D(5.40,\ 0)$ in metres measured from A, so its length is $\sqrt{1.80^2 + 2.40^2} = 3.00$ m and its direction from I is $(0.6,\ -0.8)$.

What the joint method would cost. Without the section, a joint must have at most two unknown members. Starting from the pin end, the order is A, B, H, C, I. That is **five joint free-body diagrams**, up to ten equations, and any arithmetic slip in joint A is carried through all the rest. Starting from the roller end is worse: G, F, L, E, K, J, D, which is **seven joints**. The section replaces all of this with a single equation.

5. Solution

Reactions. Moments about A (CCW positive), loads at $x = 1.80,\ 3.60,\ 5.40,\ 7.20,\ 9.00$ m and the roller at $x = 10.8$ m:

$$\sum M_A = G_y(10.8) - 2.00(1.80) - 9.00(3.60) - 11.0(5.40) - 4.00(7.20) - 3.00(9.00) = 0$$
$$G_y = \frac{3.60 + 32.4 + 59.4 + 28.8 + 27.0}{10.8} = \frac{151.2}{10.8} = 14.0 \text{ kN (up)}.$$

$$\sum F_y = A_y + 14.0 - 29.0 = 0 \ \Rightarrow\ A_y = 15.0 \text{ kN (up)},$$
$$\sum F_x = A_x = 0.$$

Section, left part. The forces on the left part are $A_y = 15.0$ kN up, 2.00 kN down at B, 9.00 kN down at C, and the three cut forces, assumed tension. A tensile F_{ID} pulls the left part toward D, so it acts along $(0.6,\ -0.8)$ and has a downward component $0.8\,F_{ID}$. The two chord forces are horizontal and drop out:

$$\sum F_y = 15.0 - 2.00 - 9.00 - 0.8\,F_{ID} = 0$$
$$4.00 = 0.8\,F_{ID} \ \Rightarrow\ F_{ID} = \frac{4.00}{0.8} = 5.00 \text{ kN}.$$

The result is positive, so the assumed tension is correct.

6. Answer

$$\boxed{F_{ID} = 5.00 \text{ kN (T)}}$$

Reactions: $A_x = 0$, $A_y = 15.0$ kN (up), $G_y = 14.0$ kN (up). Reaching ID by joints alone would have taken five joints from A (seven from G); the section took one equation.

7. Sanity check

Shear argument. The left part carries 15.0 kN up from the support and 11.0 kN down from the loads, so 4.00 kN of vertical shear must cross the panel. Only the diagonal can carry vertical force there, and $0.8 \times 5.00 = 4.00$ kN. A diagonal that slopes down toward the middle of a Pratt truss under downward loads is in tension, as found.

Right-hand part. It contains D, E, F, G and the members between them. The roller force is 14.0 kN up, the loads at D, E, F total 18.0 kN down, and a tensile F_{ID} pulls the right part toward I, along $(-0.6, +0.8)$:

$$\sum F_y = 14.0 - 18.0 + 0.8\,F_{ID} = 0 \;\Rightarrow\; F_{ID} = 5.00 \text{ kN (T)},$$

the same value from the other side.

Horizontal balance (unused equation). Moments for the two chords on the left part give independent values. About I (the cut forces IJ and ID pass through I; CD acts 2.40 m below it):

$$\sum M_I = -15.0(3.60) + 2.00(1.80) + 2.40\,F_{CD} = 0 \;\Rightarrow\; F_{CD} = 21.0 \text{ kN (T)}.$$

About D (CD and ID pass through D; IJ acts 2.40 m above it):

$$\sum M_D = -15.0(5.40) + 2.00(3.60) + 9.00(1.80) - 2.40\,F_{IJ} = 0 \;\Rightarrow\; F_{IJ} = -24.0 \text{ kN (C)}.$$

Then, on the left part,

$$\sum F_x = F_{IJ} + F_{CD} + 0.6\,F_{ID} = -24.0 + 21.0 + 3.00 = 0,$$

which uses F_{ID} and confirms it. The pattern is again beam-like: top chord squeezed, bottom chord stretched.

6 Two methods, one member: sections checked by joints

Level: exam style. Chapter 6. Target confusion: choosing the method. Use a section to get a few members quickly, then use joints from a support as an independent check on one of them.

1. Given / Find

A Warren truss spans 9.00 m. The bottom joints A, C, E, G are 3.00 m apart, and the top joints B, D, F lie 2.00 m above the bottom line at $x = 1.50$, 4.50, 7.50 m measured from A. The members are the bottom chord AC, CE, EG, the top chord BD, DF, and the diagonals AB, BC, CD, DE, EF, FG. A pin supports A and a roller supports G (vertical reaction). The loads on the top joints are: at B, 6.00 kN downward and 6.00 kN horizontally to the right; at D, 10.0 kN downward; at F, 8.00 kN downward.

Find:

1. the reactions;
2. by the method of sections with one cut at $x = 3.75$ m, the forces in BD, CD and CE;
3. an independent check of F_{CD} by the method of joints, starting at the pin A;
4. a short comparison of the two methods.

State T or C for every member force.

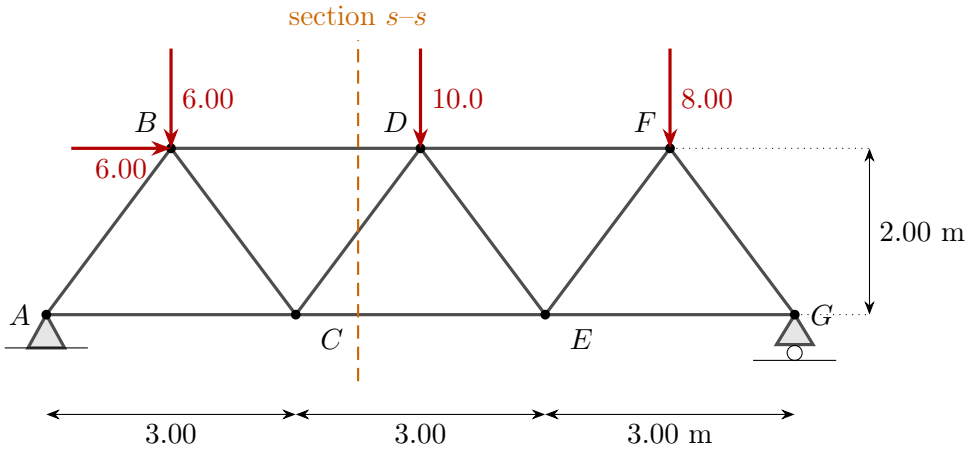


Figure 12: Warren truss with a pin at A and a roller at G. Loads in kN: 6.00 down and 6.00 to the right at B, 10.0 down at D, 8.00 down at F. The orange dashed line marks section s–s at $x = 3.75$ m, which passes through only BD, CD and CE.

2. Figure

3. Free-body diagrams

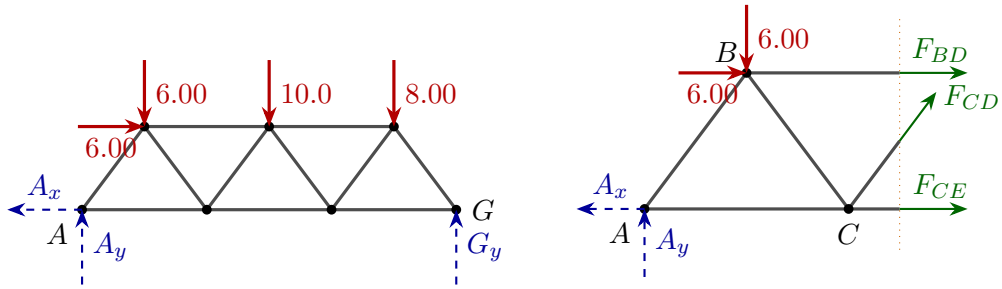


Figure 13: Left: whole-truss free-body diagram. Right: the part to the left of section s–s, containing A, B, C and members AB, AC, BC, plus the cut ends of BD, CD and CE, drawn as tension. The horizontal reaction comes out negative, so A_x acts to the left and is drawn that way (6.00 kN); $A_y = 10.0$ kN is up.

4. Strategy

Method 1: sections. Find the reactions. Then cut at $x = 3.75$ m. A vertical line at that position crosses the top chord BD ($x = 1.50$ to 4.50), the diagonal CD ($x = 3.00$ to 4.50) and the bottom chord CE ($x = 3.00$ to 6.00), and nothing else. Keep the left part, which has the support reactions and the loads at B. All diagonals run 1.50 m across and 2.00 m up over a length of 2.50 m, so their direction cosines are (0.6, 0.8). Choose the three equations to give one unknown each:

- About C: CE and CD pass through C, so only F_{BD} appears.

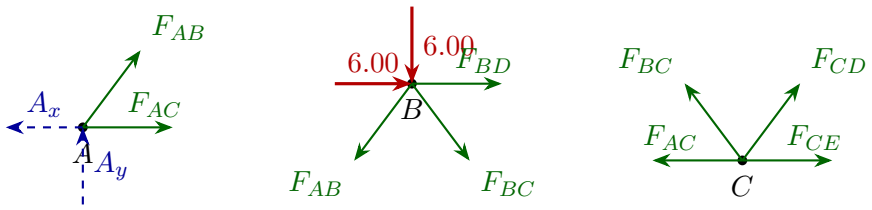


Figure 14: Joint free-body diagrams for the check by joints, in the order A, B, C. Every member force (green) is drawn pulling away from the joint, which assumes tension. Direction cosines are (0.6, 0.8) for all diagonals. A joint equation uses the value already found at the previous joint.

- About D: BD and CD pass through D, so only F_{CE} appears.
- $\sum F_y = 0$: both chords are horizontal, so only F_{CD} appears.

$\sum F_x = 0$ is kept as a check.

Method 2: joints, to verify F_{CD}. Member CD is not at the pin, so start at A (two unknowns, AB and AC), then B (with F_{AB} known, the unknowns are F_{BC} and F_{BD}), then C (with F_{AC} and F_{BC} known, the unknowns are F_{CD} and F_{CE}). Three joints reach CD. This is a good check because it uses none of the section equations.

5. Solution

Reactions. Moments about A (CCW positive). The 6.00 kN rightward load acts 2.00 m above A, so it gives a clockwise moment:

$$\sum M_A = G_y(9.00) - 6.00(1.50) - 10.0(4.50) - 8.00(7.50) - 6.00(2.00) = 0$$
$$G_y = \frac{9.00 + 45.0 + 60.0 + 12.0}{9.00} = \frac{126}{9.00} = 14.0 \text{ kN (up)}.$$

$$\sum F_y = A_y + 14.0 - 6.00 - 10.0 - 8.00 = 0 \;\Rightarrow\; A_y = 10.0 \text{ kN (up)},$$
$$\sum F_x = A_x + 6.00 = 0 \;\Rightarrow\; A_x = -6.00 \text{ kN (acts left)}.$$

Method 1, section on the left part. Forces on the left part: $A_x = 6.00$ kN to the left, $A_y = 10.0$ kN up, the two loads at B, and the cut forces, assumed tension. F_{BD} and F_{CE} pull to the right. F_{CD} pulls toward D along (0.6, 0.8).

Moments about C (3.00, 0). Here A_y is 3.00 m to the left and gives a clockwise moment. A_x acts along the line through C and gives none. At B, the downward load has lever arm 1.50 m to the left (counterclockwise), and the rightward load acts 2.00 m above C (clockwise). The top-chord force acts 2.00 m above C and pulls right, clockwise for a tensile value:

$$\sum M_C = -10.0(3.00) + 6.00(1.50) - 6.00(2.00) - F_{BD}(2.00) = 0$$
$$-30.0 + 9.00 - 12.0 = 2.00\,F_{BD} \;\Rightarrow\; F_{BD} = -16.5 \text{ kN}.$$

Moments about D (4.50, 2.00). A is 4.50 m left of and 2.00 m below D. A_y gives a clockwise moment $-10.0(4.50)$. A_x points left at a point below D, which is also clockwise: $-6.00(2.00)$. The

downward load at B is 3.00 m to the left of D (counterclockwise) and the rightward load acts along the line through D (no moment). The bottom-chord force acts 2.00 m below D and pulls right, counterclockwise for a tensile value:

$$\sum M_D = -10.0(4.50) - 6.00(2.00) + 6.00(3.00) + F_{CE}(2.00) = 0$$
$$-45.0 - 12.0 + 18.0 = -2.00\,F_{CE} \;\Rightarrow\; F_{CE} = +19.5 \text{ kN}.$$

Vertical forces. Only A_y, the downward load at B and the vertical part of F_{CD} act vertically:

$$\sum F_y = 10.0 - 6.00 + 0.8\,F_{CD} = 0 \;\Rightarrow\; F_{CD} = -\frac{4.00}{0.8} = -5.00 \text{ kN}.$$

Method 2, joints (check of F_{CD}).

Joint A. The member AB leaves along (0.6, 0.8) and AC along (1, 0):

$$\sum F_y = 10.0 + 0.8\,F_{AB} = 0 \;\Rightarrow\; F_{AB} = -12.5 \text{ kN},$$
$$\sum F_x = -6.00 + 0.6\,F_{AB} + F_{AC} = 0 \;\Rightarrow\; F_{AC} = 6.00 + 7.50 = 13.5 \text{ kN}.$$

Joint B. Directions leaving B: BA along $(-0.6, -0.8)$, BC along $(0.6, -0.8)$, BD along $(1, 0)$. The loads are 6.00 kN down and 6.00 kN to the right:

$$\sum F_y = -0.8\,F_{AB} - 0.8\,F_{BC} - 6.00 = 10.0 - 0.8\,F_{BC} - 6.00 = 0 \;\Rightarrow\; F_{BC} = 5.00 \text{ kN},$$
$$\sum F_x = -0.6\,F_{AB} + 0.6\,F_{BC} + F_{BD} + 6.00 = 7.50 + 3.00 + 6.00 + F_{BD} = 0 \;\Rightarrow\; F_{BD} = -16.5 \text{ kN}.$$

Joint C (no external load). Directions leaving C: CA along $(-1, 0)$, CB along $(-0.6, 0.8)$, CD along $(0.6, 0.8)$, CE along $(1, 0)$:

$$\sum F_y = 0.8\,F_{BC} + 0.8\,F_{CD} = 0 \;\Rightarrow\; F_{CD} = -F_{BC} = -5.00 \text{ kN},$$
$$\sum F_x = -F_{AC} - 0.6\,F_{BC} + 0.6\,F_{CD} + F_{CE} = -13.5 - 3.00 - 3.00 + F_{CE} = 0 \;\Rightarrow\; F_{CE} = 19.5 \text{ kN}.$$

6. Answer

$$\boxed{F_{BD} = 16.5 \text{ kN (C)}}, \qquad \boxed{F_{CD} = 5.00 \text{ kN (C)}}, \qquad \boxed{F_{CE} = 19.5 \text{ kN (T)}}$$

Reactions: $A_x = 6.00$ kN (left), $A_y = 10.0$ kN (up), $G_y = 14.0$ kN (up). The method of joints gives $F_{CD} = 5.00$ kN (C) independently, so the two methods agree.

7. Sanity check

The two methods agree. Joints give $F_{CD} = -5.00$ kN, the same as the section. As a bonus, the joint walk also returns $F_{BD} = -16.5$ kN (at B) and $F_{CE} = +19.5$ kN (at C), matching the other two section results. Three values matching from different equation sets is strong evidence that the reactions and the geometry are right.

Horizontal balance (unused section equation). On the left part,

$$\sum F_x = -6.00 + 6.00 + F_{BD} + F_{CE} + 0.6\,F_{CD} = 0 + (-16.5) + 19.5 - 3.00 = 0.$$

Physical sense. The top chord is squeezed and the bottom chord is stretched, as in a beam in bending. The diagonal CD is in compression. The left part has a net $10.0 - 6.00 = 4.00$ kN upward from the supports and the load at B, so the cut diagonal must supply a downward component on it. A diagonal that rises to the right from the cut can only do that by pushing, which is compression. (In the previous example the diagonal sloped down to the right and was in tension.)

Comparison of the methods.

	Sections	Joints
Work here	3 equations, 3 members	6 equations, 3 joints, 6 members
Best when	only a few members are needed, especially far from the supports	every member is needed, or the member sits next to a support
Main risk	a bad cut (more than three unknowns) or a wrong moment arm	an early slip propagates to every later joint
Strength	the answer does not depend on the other members	systematic, a joint can be checked by closing its force polygon

Use a section when the target is a handful of members deep in the truss. Use joints when you need the whole set of member forces or the target is within two or three joints of a support. On an exam, getting a member by one method and confirming it by the other, as here, is the cleanest way to catch a reaction or sign error.